1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

High stromal-to-tumor ratio (TSR), as a readout of TGF-β–driven fibroblast activation and extracellular matrix deposition, is associated with worse overall survival (OS) in breast cancer; this adverse effect is amplified when circulating HDL-C is low because reduced HDL-C fails to counteract TGF-β–mediated fibrotic and immunosuppressive signaling, promoting an immune-excluded microenvironment. Consequently, patients with high TSR and low HDL-C will have the poorest OS, identifying a subgroup that may benefit from anti–TGF-β or lipid-modulating interventions and can be stratified using AI-enabled biomarker signatures alongside the Tier 1 anchors.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	TGF-β, HDL-C
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TGF-β: TGF-β (fibrogenic/stroma) path because it mechanistically explains dense, immune-excluding stroma, and on the HDL-C (systemic lipid/metabolic) path because systemic lipid status can modulate inflammation, ECM remodeling, and immune function—together these provide a plausible interaction.

HDL-C: HDL-C), and computational/accessible biomarkers—suggesting that stromal-rich tumors do not always confer the same prognosis; the key question is what molecular conditions determine when high stromal content is prognostically adverse.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	pZVR	Supporting modifier from knowledge graph
Secondary	PTGDS	Supporting modifier from knowledge graph

Secondary	MSI	Supporting modifier from knowledge graph
Secondary	AI-based biomarkers	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: retrospective analysis in TCGA-BRCA and external cohorts to assess TSR × HDL-C interaction on OS using multivariate Cox models with robust adjustment for confounders;

Prediction 2: mediation analyses to evaluate whether TGF-β pathway activation and ECM gene signatures mediate the TSR–OS relationship;

Prediction 3: immune-infiltration profiling (e.g., CIBERSORT/xCell) to link TSR/HDL-C with immune contexture;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- pZVR: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- PTGDS: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.